

Лабораторная работа №9

Использование протокола STP. Агрегирование каналов

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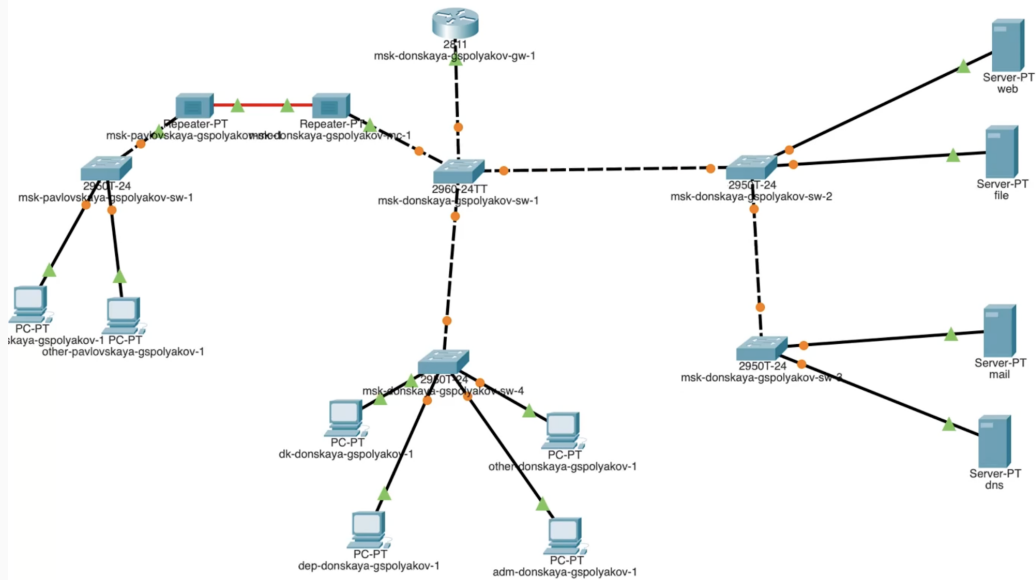
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Информация

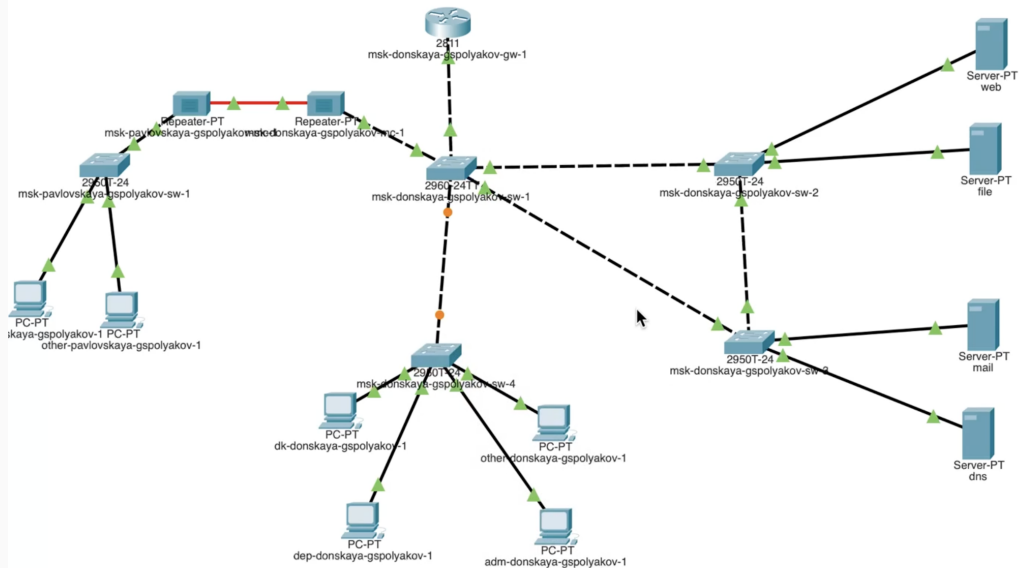
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Выполнение работы

Trunk порты



Trunk порты



Trunk порты

```
msk-donskaya-gspolyakov-sw-3>enable
Password:
msk-donskaya-gspolyakov-sw-3#configu
msk-donskaya-gspolyakov-sw-3#configure ter
msk-donskaya-gspolyakov-sw-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-gspolyakov-sw-3(config)#int g0/2
msk-donskaya-gspolyakov-sw-3(config-if)#sw
msk-donskaya-gspolyakov-sw-3(config-if)#switchport mo
msk-donskaya-gspolyakov-sw-3(config-if)#switchport mode tr
msk-donskaya-gspolyakov-sw-3(config-if)#switchport mode trunk
msk-donskaya-gspolyakov-sw-3(config-if)#exit
msk-donskaya-gspolyakov-sw-3(config)#exit
msk-donskaya-gspolyakov-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-gspolyakov-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-gspolyakov-sw-3#
```

```
C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.0.2

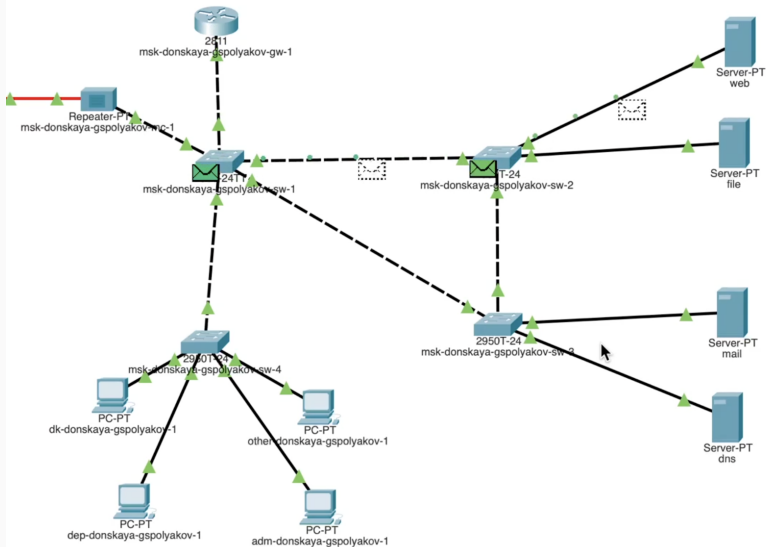
Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time=1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```


Настройка STP



```
msk-donskaya-gspolyakov-sw-2#show spanning-tree vlan 3
```

```
VLAN0003
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority      32771
             Address      0001.4347.D752
             Cost        4
             Port        26(GigabitEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID    Priority      32771 (priority 32768 sys-id-ext 3)
             Address      0001.6317.B187
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/1	Desg	FWD	4	128.25	P2p
Gi0/2	Root	FWD	4	128.26	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

```
msk-donskaya-gspolyakov-sw-2#
```

```
msk-donskaya-gspolyakov-sw-1(config)#spanning-tree vlan 3 root primary  
msk-donskaya-gspolyakov-sw-1(config)#
```

```
msk-donskaya-gspolyakov-sw-2(config)#int f0/1
msk-donskaya-gspolyakov-sw-2(config-if)#spann
msk-donskaya-gspolyakov-sw-2(config-if)#spanning-tree pro
msk-donskaya-gspolyakov-sw-2(config-if)#spanning-tree prof
msk-donskaya-gspolyakov-sw-2(config-if)#spanning-tree por
msk-donskaya-gspolyakov-sw-2(config-if)#spanning-tree portfast
```

```
msk-donskaya-gspolyakov-sw-2(config)#int f0/2  
msk-donskaya-gspolyakov-sw-2(config-if)#spanning-tree portfast
```

```
msk-donskaya-gspolyakov-sw-3(config)#int f0/1  
msk-donskaya-gspolyakov-sw-3(config-if)#spa  
msk-donskaya-gspolyakov-sw-3(config-if)#spanning-tree p  
msk-donskaya-gspolyakov-sw-3(config-if)#spanning-tree portfast
```

```
msk-donskaya-gspolyakov-sw-3(config)#int f0/2  
msk-donskaya-gspolyakov-sw-3(config-if)#spanning-tree portfast
```

```
msk-donskaya-gspolyakov-sw-2(config)#int f0/1  
msk-donskaya-gspolyakov-sw-2(config-if)#so  
msk-donskaya-gspolyakov-sw-2(config-if)#sh  
msk-donskaya-gspolyakov-sw-2(config-if)#shutdown
```



```
msk-donskaya-gspolyakov-sw-1(config)#spanning-tree mode rapid-pvst  
msk-donskaya-gspolyakov-sw-1(config)#
```

```
msk-donskaya-gspolyakov-sw-2(config)#spanning-tree mode rapid-pvst  
msk-donskaya-gspolyakov-sw-2(config) #
```

```
msk-donskaya-gspolyakov-sw-3(config)#spanning-tree mode rapid-pvst  
msk-donskaya-gspolyakov-sw-3(config)#
```

```
msk-donskaya-gspolyakov-sw-4(config)#spanning-tree mode rapid-pvst  
msk-donskaya-gspolyakov-sw-4(config) #
```

```
msk-pavlovskaya-gspolyakov-sw-1(config)#spanning-tree mode rapid-pvst  
msk-pavlovskaya-gspolyakov-sw-1(config)#
```

Агрегирование каналов

```
C:\>ping -n 1000 mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=31ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

Агрегирование каналов

